

Fraud Risk Scoring

Ranking transactions for limited review capacity

DATA / VALIDATION

Simulated transaction data

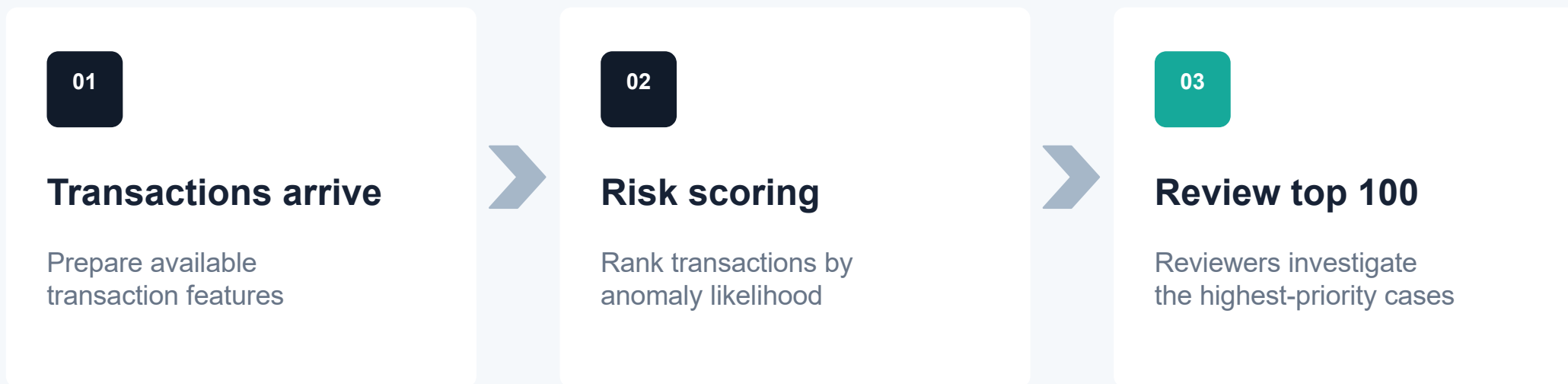
Time split · 7-day label delay

**Final test: 66,741 transactions
614 simulated anomalies**

Not production performance or real customer behavior.

With capacity for 100 reviews a day, which transactions come first?

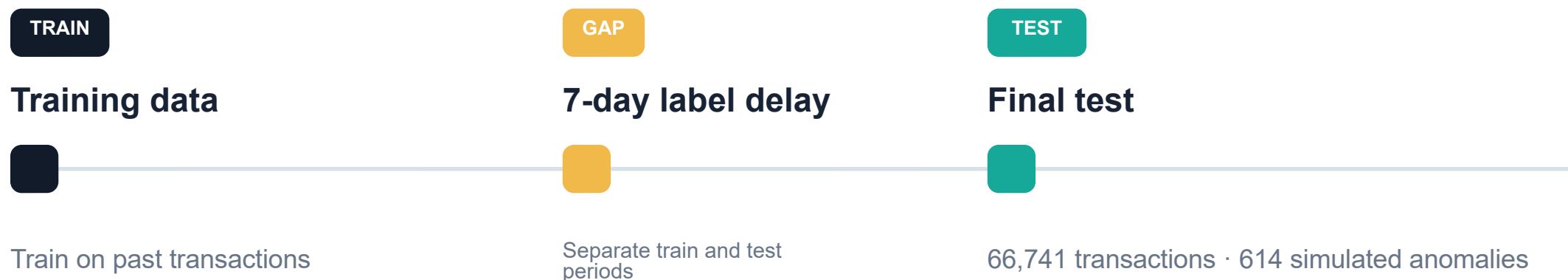
The goal is not to automatically decide every transaction, but to prioritize review.



Decision unit = top 100 reviews · Evaluate PR-AUC, Precision@100, and Recall@100 together

Validated in chronological order to avoid using future information

A 7-day label-delay gap separates training from final evaluation to reduce leakage risk.

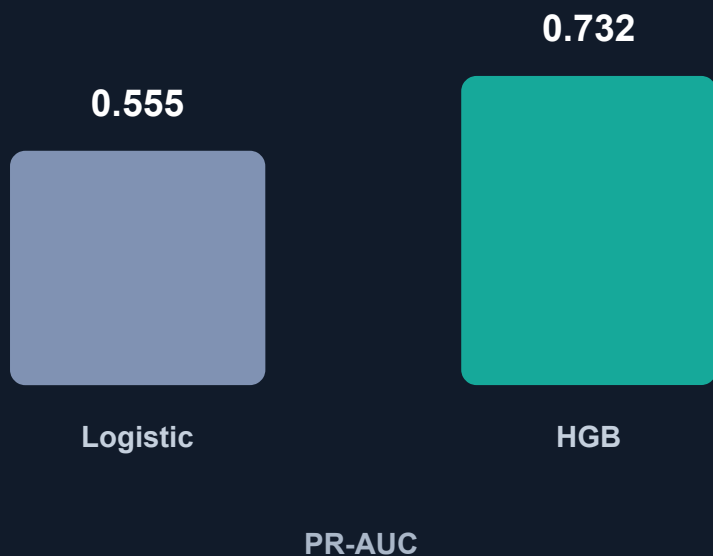


Validation checks

Time split · 7-day gap · Leakage control · Review-capacity metrics

HGB captures more simulated anomalies at the same review capacity

Compared HistGradientBoosting (HGB) against a Logistic Regression baseline.



Top-100 review queue

Precision@100

63.7%

Share of anomalies in the 100 reviewed cases

Recall@100

72.6%

Share of all anomalies included in the top 100

The 50–99 amount range is a priority segment for review

Segment analysis for the 100-review simulation; not generalizable to real customer behavior.

Amount range: 50–99

58.8%

Recall



Captured: 94

Missed: 66

What was missed

**Simulated anomalies
that remained outside
the top-100 review
queue**

How to interpret it

**Not a deployment
decision.
Priority segment for
feature
and threshold review**

Separate what was validated from what remains before real deployment

These are offline prototype results on public simulated data.

Validated in this project

- Time split and 7-day label delay
- Leakage control
- Performance vs. baseline
- Segment differences at 100 reviews/day

Still needed before real deployment

- Representativeness of real customers/transactions
- Loss reduction and customer experience
- Fairness and regulatory suitability
- Online serving and monitoring performance

Next: enrich weak-segment features → revalidate → review operations and fairness